



International School

CAPSTONE PROJECT 1

CMU-SE 450

PROJECT PLAN DOCUMENT

Date: September 14, 2026

**A Personalized GraphRAG Assistant for Exploring, Understanding and
Planning Local Cultural Experiences**

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PROJECT INFORMATION

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1. Introduction

1.1 Purpose

This document provides a summary of the project's objectives, assignments, key milestones, required resources, timelines, overall schedule and budget allocation used based on the Proposal to build a Personalized GraphRAG Assistant platform that supports the discovery, in-depth understanding and planning of local cultural experiences in Hue and Da Nang on time, according to requirements and plans.

1.2 Project overview & technology used

HeritageGraph (CMG) is an AI assistant platform that assists in discovering, deeply understanding, and planning local cultural experiences in Hue and Da Nang, covering 6 main cultural categories: Historical Sites, Culinary, Scenic Spots, Performing Arts, Festivals, and Traditional Craft Villages. The system combines Hybrid RAG technology (BM25 keyword channel, pgvector vector channel, and unsigned n-gram channel) with GraphRAG Knowledge Graph to provide accurate answers with source citations to each sentence and abstention gates when evidence is missing

Table 1. Technology Stack

Architectural Floor	Ingredients	Applicable technology
Frontend	Framework & UI	Next.js 14 App Router, React 18, TypeScript, Ant Design
Frontend	Maps & 3D Viewer	Leaflet with OpenStreetMap, Three.js built-in <model-viewer> (Draco/Meshopt compression)
Backend	API & Servers	Python, FastAPI, Pydantic v2, Uvicorn
Data storage	Databases & Vector Stores	PostgreSQL 16 with SQLAlchemy, Alembic, and pgvector extensions

The project uses a locally fine-tuned Vietnamese LLM / Adapter Checkpoint to support the creation of responses based on proven data sources. The model is integrated with the Hybrid RAG and GraphRAG pipelines to generate well-grounded, cited, and potentially rejected answers when the system does not have enough reliable evidence.

The model supports the main functions:

- **Grounded Narration:** create content based on information retrieved from the system's data source.
- **Citation:** provides a source citation for the answer so that the user can verify the information.
- **Abstention:** refusing to answer when the system does not have enough reliable evidence.
- **False-Premise Correction:** detects and handles questions that contain false assumptions, limiting the generation of inaccurate information.
- **Vietnamese Cultural Context:** create Vietnamese content that is appropriate for the local cultural context.

Local fine-tuned Vietnamese LLM / Adapter Checkpoint is seen as a technical deliverable of the project, combined with Hybrid RAG and GraphRAG to provide grounded, cited and capable rejection answers in the absence of evidence.

1.3 Project handover products

The project is implemented to serve two main groups of users:

- **Admin / Culture Manager:** Manages cultural documents, structured records (Places, Events, Artifacts) with source checks, browses metadata/alternate names, moderates AI-derived entities and relationships, views audit logs, and monitors graph health on the Admin Dashboard.
- **End-users (Students, Teachers, Researchers, Communities):** Explore cultures by category and region, Q&A in Vietnamese with GraphRAG Chatbot with sentence-by-sentence source citations and the ability to reject when data is missing, receive personalized suggestions according to explainable graph paths, view proactive schedule/weather advisor cards, and experience a detailed page in the form of a Tapestry integrated with maps, timelines, 3D viewers (.glb) with bilingual audio commentary (Vietnamese - English).

2. Team organization

2.1 Group Information

Table 2. Team organization

Full Name	Tel	Email	Role
Mcs. Nguyen Thi Thanh Tam	0364130007	ttamdtu@gmail.com	Project Supervisor
Duong Thanh Hung	0905001371	hungthanhhung37@gmail.com	Project Leader/Full-stack Developer, QA
Cao Van Khoa	0355737634	caovankhoa11@gmail.com	Data and Knowledge Engineer
Dao Ho Anh Tung	0345609705	daohoanhtung@gmail.com	UI/UX and Frontend Developer
Le Anh Nghia	0398539477	leanhngghia23@gmail.com	UI/UX and Frontend Developer
Duong Bao Ngoc	0773020847	ngocduong.200011@gmail.com	QA and Documentation

2.2 Roles and Responsibilities

Table 3. Roles and Responsibilities

Role	Responsibilities	Name / Title
Project Leader / Full-stack Developer, QA	Monitoring project progress and scope; risk management; coordinating Agile/Scrum workflows; designing the overall system architecture; developing REST API (FastAPI) and GraphRAG Pipeline orchestration; integrating sentence citation verification modules and abstention gates; managing system-wide quality testing.	Duong Thanh Hung
Data and Knowledge Engineer	Collecting, standardizing and processing cultural archives (80+ documents in 6	Cao Van Khoa

	categories); designing Knowledge Graph (NetworkX) and Vector Index schemas (PostgreSQL 16 with pgvector); building pipelines extracting entities, relationships, aliases, and structured cultural records (Places, Events, Artifacts) with source checking.	
UI/UX and Frontend Developer	Design and develop web user interfaces (Next.js 14, React 18, TypeScript, Ant Design) for screens: Card-based Chatbot, Tapestry-style Entity Detail Page, Map Frame (Leaflet/OpenStreetMap), Timeline, 3D Model Viewer (Three.js/.glb) with integrated bilingual voiceover audio player (Vietnamese - English).	Dao Ho Anh Tung
UI/UX and Frontend Developer	Design and development of the Admin Dashboard interface: Forms for managing and uploading cultural documents/records, AI-extracted entity and relationship moderation/verification screens, personalized interest profile management interfaces, and audit log displays.	Le Anh Nghia
QA and Documentation	Develop a Test Plan; design and execute test scenarios (Unit, Integration, E2E); quantitatively evaluate AI indicators (Precision@5, Recall@1, Citation Faithfulness, Citation Coverage, Abstention Accuracy, Latency p95); compile SRS documents, User Guides, technical reports and project acceptance documents.	Duong Bao Ngoc
Project Supervisor (Mentor)	Orient the project implementation process; monitor and check progress on a weekly basis; review and evaluate software & AI	MSc. Nguyen Thi Thanh Tam

	architecture decisions; approve end-of-term deliverables.	
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2.3 Communication methodology

Table 4. Communication Methodology

Attendees	Theme / Product Handover	Frequency	Method
Instructors and Team Members	Project Progress Review & Direction Assessment	Weekly	Meet in person, Email. Zalo
Team Members	Progress Review & Quick Meeting	Daily/Weekly	Google Meet

2.4. Communication and Reporting

Table 5. Communication and Reporting

Type of communication	Methods and tools	Frequency	Information	Human Resources
Requirements & Data Analysis Meetings	Meet in person, Google Meet	Periodic during the Initiation phase	Finalize the Requirements Specification (SRS), close the cultural data repository (at least 80 documents in 6 categories in Hue & Da Nang and Place, Event, Artifact records with source citations), clarify the scope of SRS.	Project Team
Design & Architecture Review Meeting	Google Meet, Figma, Jira	1 time/week	Review and finalize UI/UX (Card Chat, Antd-style Detail Page, 3D viewer), PostgreSQL/pgvector Knowledge Graph schema, and Hybrid RAG + GraphRAG pipeline architecture.	Project Team
Deployment Checkpoints	Discord, GitHub, Jira	Weekly	Track the development progress of Backend FastAPI, PostgreSQL/pgvector, Frontend Next.js, Personalization modules, and Proactive Advisors according to each Sprint milestone.	Project Team & Instructor
Test Feedback & AI Reviews	Meet in person, Google Meet	After each test cycle	Share test results, quantitatively evaluate AI indicators (Recall@1 Retrieval Accuracy, Citation Faithfulness/Coverage, Abstention Accuracy, p95 Latency).	Project Team
Handover & Acceptance	Meet in person, Google Meet	Once at the end	Demonstration of the final Web Prototype	Project Team & Instructor

Meeting			product (including 2 experience itineraries Research & Tour, 3D Viewer & Audio bilingual narration), handover of all technical documents and summary reports.	
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3. Development process

The project applies an Agile/Scrum agile development methodology with short one-week iterations. Each sprint includes activities: planning, development, testing, live demo with instructors, and retrospective meetings. The development process adheres to two core principles: prioritizing addressing high-risk items first (data authoring, baseline latency measurement) and not building a back-tier on an unmeasured evaluative floor.

Table 6. Development process

Stage	Time	Contents of implementation
Initiation	Weeks 1-2	Problem Analysis, Scope, Proposal, SRS, Data Dictionary, Wireframe, Evaluation Plan, PostgreSQL Schema
Sprint 1: Foundation, Image & Document Ingestion	Weeks 3-5	Account, consent, data management, image upload, metadata, Gallery, Image Story và document ingestion
Sprint 2: Knowledge Base, Extraction & Personalization	Weeks 6-8	80+ documents, extraction, evidence, aliases, structured records, vector index, Knowledge Graph và personalization
Sprint 3: Retrieval GraphRAG, Answering & End-to-End Experience	Weeks 9-11	Hybrid RAG, GraphRAG, Local fine-tuned Vietnamese LLM, citation, abstention, proactive advisory, interface, multimedia và 2 End-to-End Journeys
Testing	Weeks 12-13	Integration, E2E, performance, correctness, AI metrics và hallucination evaluation
Closing	Weeks 14-15	Tuning, privacy/ethics review, documentation, evaluation report, demo video và presentation.

4. Schedule and cost

4.1 Detailed schedule

Table 7. Details of the project schedule

Total duration: 15 weeks, from 27/08/2026 to 06/12/2026

STT	Job Name	Start Date	End Date	Hours
1	Initiation – Startup & Analysis	27/08/2026	09/09/2026	160 hours
1.1	Problem analysis, data source survey and scoping Hue & Da Nang	27/08/2026	29/08/2026	40 hours
1.2	Define Requirements, Design Data Dictionary, Schema, and Wireframe	30/08/2026	04/09/2026	56 hours
1.3	Finalize Proposals, SRS, Evaluation Plans, and initialize PostgreSQL Databases using Docker Compose	05/09/2026	09/09/2026	64 hours
2	Sprint 1 – Foundation Image & Document Ingestion	10/09/2026	30/09/2026	240 hours
2.1	PostgreSQL database building, migration, Docker Compose, registration/login, and session management	10/09/2026	16/09/2026	48 hours
2.2	Consent logging, personal-data export/delete, and baseline latency measurement	17/09/2026	19/09/2026	24 hours
2.3	Image upload, metadata, Gallery và Image Story	20/09/2026	24/09/2026	56 hours
2.4	Document ingestion for TXT/DOCX, Unicode standardization, and passages generation	25/09/2026	27/09/2026	48 hours
2.5	Finalizing the Design System and testing Sprint 1	28/09/2026	30/09/2026	64 hours

3	Sprint 2 – Knowledge Base Extraction & Personalization	01/10/2026	21/10/2026	240 hours
3.1	Expand corpus to a minimum of 80+ documents in 6 cultural categories	01/10/2026	05/10/2026	48 hours
3.2	Extract Entity, Relation, Alternate Names/Aliases, and Evidence	06/10/2026	09/10/2026	48 hours
3.3	Build Place, Event, Artifact records with Source Check and Provenance	10/10/2026	13/10/2026	40 hours
3.4	Build a Knowledge Graph, Vector Index, and Hybrid Retrieval baseline	14/10/2026	16/10/2026	40 hours
3.5	Develop implicit preference profiles, time-based decay, and Profile View/Edit	17/10/2026	19/10/2026	32 hours
3.6	Xây dựng Explainable Graph-Path Recommender, cross-category bonus và diversity re-ranking	20/10/2026	21/10/2026	32 hours
4	Sprint 3 – Retrieval GraphRAG, Answering & End-to-End Experience	22/10/2026	11/11/2026	240 hours
4.1	Hybrid Retrieval Development: BM25 + Vector/pgvector + unsigned n-gram	22/10/2026	25/10/2026	40 hours
4.2	Development and refinement of the Local fine-tuned Vietnamese LLM/Adapter Checkpoint	26/10/2026	28/10/2026	32 hours
4.3	Build GraphRAG re-ranking, trace endpoints, and citation	29/10/2026	31/10/2026	32 hours

	checks			
4.4	Xây dựng abstention gates, false-premise correction và grounded Vietnamese answering	01/11/2026	02/11/2026	32 hours
4.5	Built-in 6-class intent classifier, Proactive Advisory Cards and Weather API integration	03/11/2026	05/11/2026	40 hours
4.6	Complete Card Chat, Tapestry-style Detail, Timeline, Map, 3D Viewer and bilingual Audio	06/11/2026	08/11/2026	32 hours
4.7	Complete 2 End-to-End Journeys: Research Journey and Attendance Journey	09/11/2026	11/11/2026	32 hours
5	Testing & Evaluation	12/11/2026	25/11/2026	160 hours
5.1	Unit, Integration, API, UI and E2E Testing	12/11/2026	18/11/2026	80 hours
5.2	Retrieval Assessment, Entity F1, Citation Faithfulness/Coverage, Abstention, Intent, Recommendation, and p95 Latency	19/11/2026	25/11/2026	80 hours
6	Wrap-up – Finishing & Handing over	26/11/2026	06/12/2026	160 hours
6.1	Tuning, Privacy/Ethics Review, Technical Documentation and API Specs	26/11/2026	30/11/2026	64 hours
6.2	Complete User Guide, Evaluation Report and Demo Video	01/12/2026	03/12/2026	48 hours
6.3	Finishing the Slide, Commissioning and Preparing Capstone Protection 1	04/12/2026	06/12/2026	48 hours

4.2. Cost

4.2.1 Cost of Working Hours / Person

Table 8. Cost of working hours / person

Full Name	Role	Salary (USD/hour)
Duong Thanh Hung	Project Lead, Backend & AI Programmer	4
Cao Van Khoa	Data & Knowledge Engineer	4
Dao Ho Anh Tung	Frontend Programmer (User Interface)	4
Le Anh Nghia	Frontend Programmer (Admin Interface)	4
Duong Bao Ngoc	Quality Management (QA) & Documentation	4

4.2.2. Estimate the total cost**Table 9. Estimate of total cost**

STT	Criteria	Total cost (USD)
1	Cost of working hours	4,800
2	Other Costs (Hosting, API Credits, Domain Names)	500
Total	Total Cost	5,300

Table 10. Detailed Description

Description	Quantity	Unit
Number of Members	5	People
Total Expected Effort	1200	Hours
Hours worked per day/person	16	Hours
Number of working weeks	15	Week
Total working hours per member	240	Hours/person

❖ Explanation

Total project effort according to the detailed schedule: 1200 hours

. Labor cost: 1200 hours × \$4/hour = \$4,800.

Estimated cost of AI, hosting, and hosting services: \$500.

Total projected cost: 4,800 + 500 = \$5,300.

5. Project Risks**Table 11. Worth the feasibility and severity of each risk**

Review Code	Description
L	Rated at Low
M	Rated as Medium

H	Rated High
E	Rated as Extreme (Impact only)

Table 12. Project Risk List & Assessment

Risks	Definition	Impact	Likelihood	Mitigation strategies
Cultural data is uneven across categories	The number of encyclopedic materials is uneven between categories (e.g., historical relics are very deeply documented, but Festivals, Craft Villages and Performing Arts are sparse).	H	M	Plan to collect and manually edit data early; proactively balance the document inventory across all 6 cultural categories (minimum 8 documents/category) right in Sprint 2.
Wrong extraction of Vietnamese proper names/unaccented words	The names of relics, celebrities, artisans or Vietnamese famous place names are extracted incorrectly, or users enter unsigned questions, affecting the retrieval accuracy.	H	H	Applying a closed vocabulary, building a dictionary of alternate names/aliases, combining the 4-character n-gram retrieval channel independent of Vietnamese accents and creating a moderated Admin interface.
Psychedelic LLM/Answer No Citation	The hallucination language model is not included in the local cultural database or provides feedback without citing a verifiable source.	H	M	Prompt constraints only to reply from retrieved contexts; enforces correct citations to the sentence level [Source: <sentence> - <url>] and activates abstention gates when the similarity is below the threshold
3D data processing latency and Audio	Creating live audio voiceover files or loading	M	M	pre-generate and pre-validate TTS offline audio scripts;

Voiceover synthesis	large 3D models at the time of the request extends the system response time beyond the NFR01 threshold (p95 latency $\leq 8s$)			reuse of public open 3D models; optimal storage on Cloudflare R2 and Azure Blob Storage.
Schedule delays	Complex integration workloads between Hybrid RAG, Knowledge Graph, Personalization, and Multimedia result in the risk of not completing 15 weeks on time.	H	H	Strictly apply the Agile/Scrum process in a 1-week cycle; closely monitor the progress of timelines; only limit the 3D feature to the digitized location set instead of scanning the whole thing.

6. Product delivered

Table 13. Handover products

STT	Product Deliveries	Description
1	Web Prototype & Source Code	Full application source code with Docker Compose environment encapsulation configuration
2	Cultural Records & Data Repository	80+ standardized documents and verified sets of Place, Event, Artifact records cite sources.
3	Pipeline Hybrid RAG & GraphRAG	The 3-channel retrieval system integrates trace endpoints and measurement reports of evaluation indicators.
4	Personalization & Suggestion Module	Inference of implicit interest profiles over time with a set of Explainable Graph-Path Recommenders.
5	Local Fine-tuned Vietnamese LLM / Adapter Checkpoint	The Vietnamese model is locally fine-tuned, integrated with Hybrid RAG and GraphRAG to generate grounded answers, cite sources, support Abstention, False-Premise Correction, and Vietnamese cultural context.
6	Proactive Advisory System	The 6-type intent identifier renders the correct contextual information cards.
7	Multimedia Experience	Tapestry-style detail pages, interactive maps, timelines, and a 3D viewer Three.js built-in bilingual audio voiceovers (Vietnamese - English).
8	Two End-to-End Experience Processes	Completing two main streams: Research Journey (artifacts → place to store → artifacts of the same period → related craft villages) and Attendance Journey (festivals → calendars → places → weather → preparing → observation points → parking lots).
9	Admin Dashboard & Review Suite	Content moderation interface, audit log, and automated test reports.

The project delivers a complete Web Prototype of the HeritageGraph system and functional modules: Document & Structured Record Management with Source Check, Pipeline Hybrid RAG (BM25 + pgvector + unaccented n-gram) combined with GraphRAG Re-ranking, Vietnamese Q&A Chatbot with Abstention Gates Citations, Explainable Graph-Path Recommender, Proactive Advisory Cards, Tapestry-style Entity Detail Page with Maps, Timeline & 3D Viewer (.glb on Cloudflare R2) with Bilingual Audio Commentary (on Azure Blob Storage), and Moderation Admin Dashboard. In addition to the software product, the project handed over the Evaluation Suite, AI Evaluation Report & Performance Measurement (Evaluation Report), Technical Documents (SRS, API Specs, Data Dictionary), Test Documents and Project Summary Report.

7. References

7.1. Standards and technical documents

1. **IEEE 830 / ISO/IEC/IEEE 29148** – *Software Requirements Specification / Software Requirements Engineering*.
2. **IEEE 1016** – *Software Design Description*.
3. **IEEE 829 / ISO/IEC/IEEE 29119** – *Software Testing*.
4. **ISO/IEC/IEEE 12207:2017** – *Software Life Cycle Processes*.
5. **ISO/IEC 25010** – *Systems and Software Quality Requirements and Evaluation (SQuaRE)*.

6. **ISO/IEC 23894** – *Information Technology — Artificial Intelligence — Guidance on Risk Management*.

7.2. Documentation on AI, RAG, and GraphRAG

1. **Lewis, P., et al. (2020)**. *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*. NeurIPS 2020.
<https://arxiv.org/abs/2005.11401>
2. **Microsoft Research**. *Project GraphRAG: LLM-Derived Knowledge Graphs*.
[Microsoft Research – Project GraphRAG](#)

7.3. Documents on digital cultural heritage

1. **CyArk**. *Digital Heritage Preservation*.